

PHODZO NAGANA

Software Developer

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PROFESSIONAL SUMMARY

Motivated Software Developer with a BSc in Information Technology and over two years of hands-on experience building full-stack, enterprise, and AI-integrated applications. Proficient in Java, Spring Boot, Angular, React, Python, and SQL, with a solid understanding of the full Software Development Life Cycle (SDLC). Experienced in designing REST APIs, working with relational and non-relational databases, and deploying applications on cloud platforms including AWS, Azure, and GCP. Backed by industry certifications and a track record of delivering scalable, real-world solutions in Agile team environments.

TECHNICAL SKILLS

Programming Languages: Java, Python, C#, JavaScript, TypeScript, PHP, Visual Basic, HTML/CSS, Flutter
Frameworks & Libraries: Spring Boot, Angular, React.js, Node.js, ASP.NET Web API, .NET Framework, Flask, FastAPI, AngularJS
Databases: MySQL, Microsoft SQL Server, PostgreSQL, SQLite, MongoDB, JDBC
Cloud & DevOps: AWS, Azure, GCP, Docker, Linux, Git, GitHub, GitLab, GitFlow
Tools & Platforms: Camunda BPMN, n8n, Swagger, Open API, REST APIs, RESTful Web Services, JWT, Power BI, Excel
Methodologies: Agile/Scrum, OOP, SDLC, TDD, Business Process Management, CI/CD principles, Database Administration

WORK EXPERIENCE

Software Developer Intern First National Bank

March 2025 – March 2026
Sandton, Gauteng

- Developed and maintained backend services using Java and Spring Boot, building and documenting REST APIs to support core banking operations and internal applications.
- Designed, monitored, and improved business process workflows using the Camunda Process Engine, ensuring operational transparency and execution accuracy across enterprise systems.
- Built Angular-based frontend interfaces that presented real-time business process data to stakeholders, bridging backend services with intuitive UI.
- Wrote and optimised SQL queries to analyse, manipulate, and manage data across operational databases, ensuring efficient and accurate data retrieval.
- Won FNB's internal AI Innovation competition by engineering an AI Guardrail System that detects prompt injection attacks through a series of automated pipelines and firewalls, integrating tools including n8n.
- Collaborated with cross-functional teams in an Agile environment, contributing to sprint planning, code reviews, and iterative feature delivery.

Java Developer Intern CAPACITI

November 2024 – February 2025
Johannesburg, Gauteng

- Designed, developed, and maintained applications using Java, Spring Boot, and Node.js, applying OOP principles to build modular and maintainable code.
- Built responsive web interfaces with HTML, CSS, JavaScript, and React, ensuring cross-browser compatibility and clean user experiences.
- Worked in Linux and Git environments for source control, deployment, and structured collaborative development using GitFlow.
- Participated in Agile ceremonies including standups, sprint reviews, and retrospectives, contributing to a team-oriented delivery process.

Full Stack Developer Intern Bidvest Protea Coin

March 2024 – October 2024
Centurion, Gauteng

- Developed full-stack web applications using ReactJS, Node.js, and ASP.NET Web API, integrating multiple database systems including MongoDB, Firebase, SQLite, and Microsoft SQL Server.
- Secured applications using JSON Web Tokens (JWT) for robust authentication and role-based authorisation.
- Managed database operations and reporting using SQLite and Microsoft SQL Server, maintaining data integrity across application layers.
- Tested and debugged REST API endpoints, resolving integration issues between frontend and backend systems to ensure end-to-end reliability.

EDUCATION

Bachelor of Science in Information Technology (Software Development) *January 2021 – December 2023*

Richfield Graduate Institute of Technology

Centurion

- Key Modules: Programming, Software Engineering, Web Development, Information Systems, Cloud Computing, Artificial Intelligence, Cybersecurity

PROJECTS

TrailCast – Real-Time Weather Navigation Engine | *Java 17, Spring Boot, React, PostgreSQL/PostGIS, WebSockets, Mapbox API, OpenWeatherMap API*

- Engineered a real-time, weather-dependent navigation engine that calculates optimal routes for Driving, Off-roading, and Hiking modes based on live meteorological data.
- Architected a Spring Boot backend with PostGIS for advanced geospatial queries, detecting route intersections with active flood zones and severe weather fronts.
- Implemented a WebSocket-based alerting system that pushes live safety notifications and rerouting suggestions when weather conditions deteriorate mid-trip.
- Developed a responsive React frontend with a mobile-first UI, integrating Mapbox GL JS and OpenWeatherMap APIs for high-fidelity mapping and atmospheric intelligence.

Listings CRUD Application | *C#, ASP.NET Core MVC, Entity Framework Core, SQL Server, Bootstrap, Razor Views*

- Engineered a full-stack web application with role-based access — public users can browse and search listings while admins have full CRUD capabilities.
- Designed a robust ASP.NET Core MVC backend using Entity Framework Core for data access and SQL Server for persistent storage, with search filtering by title, description, and type.

AI Guardrail System – Prompt Injection Detection | *n8n, Python, API Integrations*

- Built and won FNB's AI Innovation competition with an intelligent pipeline system that detects and filters prompt injection attacks across multi-stage firewalls using n8n automation and API integrations.

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900)
- Oracle: Java Foundations
- Amazon: Full Stack Web Development, Generative AI in Software Development, Application Development, Programming with Java, Introduction to Software Development
- Amazon: Database Management with Java and SQL, Data Structures and Algorithms
- JUnit and Mockito Unit Testing for Java Developers
- IBM: Hands-on Introduction to Linux Commands and Shell Scripting (V1 & V2)
- Fortinet Certified Fundamentals in Cybersecurity
- Cisco: Introduction to Cybersecurity
- Coursera: Junior Software Developer

REFERENCES

Available upon request.